GPV Identification: Recovering Values from Bids

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Part I

From Theory to Data

Recap: The First-Price Auction

n bidders, values V_i i.i.d. from F over $[0, \overline{v}]$.

Symmetric Bayes-Nash Equilibrium: every bidder bids $b = \beta(v)$, with $\beta(\cdot)$ strictly increasing.

From Lecture 3:

$$\beta(v) = v - \underbrace{\frac{\int_0^v F^{n-1}(z)\,dz}{F^{n-1}(v)}}_{\text{shading} > 0}.$$

The Empirical Auction Problem

Theory (forward direction):

$$n \text{ and } F(\cdot) \implies \beta(\cdot) \implies \text{bid distribution.}$$

Empirical auctions (reverse direction):

$$\text{observe bids} \implies \text{recover } F(\cdot).$$

Why we care: predicting revenue and bidder surplus under a counterfactual mechanism (a different reserve price, a different n, a different auction format) requires knowing F, not just β.

What Is Observed, What Is Not

- Observed: n, and bids b_{it} for bidders $i = 1, \ldots, n$ in auctions $t = 1, \ldots, T$.
- Unobserved: values v_{it}, the value distribution F, and the bidding strategy β.

Caution: the empirical distribution of bids is not an estimate of F.

Bidders shade their bids, $\beta(v) < v$, so bids are systematically lower than values.

Part II

The GPV Inversion Formula

Key Idea

Guerre, Perrigne, and Vuong (2000): recover v_i from b_i using only the bidder's first-order condition, **without ever solving for β or F**.

The trick: write the first-order condition in terms of objects that are directly observable from bid data.

The Observed Bid Distribution

In a symmetric equilibrium, every bidder's bid is a draw from the same distribution.

Let $G(\cdot)$ denote the cdf of a bidder's equilibrium bid, and $g(\cdot) = G'(\cdot)$ its density.

Since $b = \beta(v)$ with β strictly increasing,

$$G(b) = \Pr(\beta(V) \leq b) = \Pr(V \leq \beta^{-1}(b)) = F(\beta^{-1}(b)),$$

or equivalently,

$$G(\beta(v)) = F(v).$$

Winning Probability (In Terms of Bids)

Bidding b wins $\iff$ b exceeds all $n-1$ rival bids:

$$\Pr(\text{win} \mid b) = G(b)^{n-1}.$$

since $G(b) = \Pr(\text{bid} \leq b)$, i.e., a (rival's) bid is below my bid b.

Optimization (In Terms of Bids)

A bidder with value v chooses b to maximize expected payoff:

$$\max_b \underbrace{(v-b)}_{\text{markup}} \cdot \underbrace{G(b)^{n-1}}_{\Pr(\text{win}|b)}.$$

First-order condition:

$$-G(b)^{n-1} + (v-b)(n-1)G(b)^{n-2}g(b) = 0.$$

The Inversion Formula

Solving the first-order condition for v:

$$\boxed{v = b + \frac{G(b)}{(n-1)\, g(b)}}$$

The right-hand side depends only on b, G, and g — all estimable directly from observed bids.

Neither F nor β appears: this is what makes F nonparametrically identified from bid data alone.

A Worked Example: $n = 2$, $V \sim \text{Uniform}[0, 1]$

Forward (values $\to$ bids). With $F(v) = v$ and $n = 2$,

$$\beta(v) = v - \frac{\int_0^v F(z)\,dz}{F(v)} = v - \frac{v^2/2}{v} = \frac{v}{2}.$$

The bid distribution. Since $b = \frac{1}{2}v$,

$$G(b) = \Pr(V \le 2b) = 2b, \qquad g(b) = 2, \qquad b \in [0, \tfrac{1}{2}].$$

So the bids are $\text{Uniform}[0, \frac{1}{2}]$ — not $\text{Uniform}[0, 1]$.

Backward (invert the bid).

$$v = b + \frac{G(b)}{(n-1)\,g(b)} = b + \frac{2b}{1 \cdot 2} = 2b.$$

The inversion returns $2b = v$ exactly, and the pseudo-values $2b$ are $\text{Uniform}[0, 1]$ — the true F recovered. The loop closes.

Part III

Nonparametric Estimation

The Two-Step Estimator

1. Estimate $\widehat{G}$ and $\widehat{g}$ nonparametrically from observed bids.
2. Invert each observed bid into a pseudo-value $\widehat{v}$ using the boxed formula.

$$\boxed{\widehat{v} = b + \frac{\widehat{G}(b)}{(n-1)\,\widehat{g}(b)}}$$

3. Estimate F (or its density f) from the pseudo-values $\widehat{v}$.

No parametric assumption on F or G is needed at any step.

Step 1: Estimating G and g

Pool the nT observed bids $\{b_{it}\}$:

$$\widehat{G}(b) = \frac{1}{nT} \sum_{i,t} \mathbf{1}\{b_{it} \leq b\}, \qquad \widehat{g}(b) = \frac{1}{nT\eta} \sum_{i,t} K\left(\frac{b - b_{it}}{\eta}\right),$$

where $K(\cdot)$ is a kernel and $\eta \to 0$ is a bandwidth.

Step 2: Pseudo-Values

For each observed bid b_{it}, compute

$$b_{it} + \frac{\widehat{G}(b_{it})}{(n-1)\,\widehat{g}(b_{it})} = \widehat{v}_{it}.$$

Treat $\{\widehat{v}_{it}\}_{i,t}$ as if it were a sample of nT i.i.d. draws from F.

Step 3: Estimating F

From the pseudo-values $\{\widehat{v}_{it}\}$, estimate the value distribution directly — exactly as you estimated G and g from bids in Step 1:

$$\widehat{F}(v) = \frac{1}{nT}\sum_{i,t} \mathbf{1}\{\widehat{v}_{it} \le v\} \quad \text{(empirical cdf: your } \texttt{my_ecdf}\text{)},$$

$$\widehat{f}(v) = \frac{1}{nT\eta'}\sum_{i,t} K\left(\frac{v - \widehat{v}_{it}}{\eta'}\right) \quad \text{(kernel density: your } \texttt{my_epdf}\text{)}.$$

$\widehat{F}$ is the nonparametric estimate of the underlying value distribution. (A smooth alternative is $\widehat{F}(v) = \int_0^v \widehat{f}(x)\,dx$; we use the empirical cdf here, to match the ECDF-for-F, kde-for-f convention you code in the homework.)

Trimming and Boundary Bias

$\widehat{g}(b)$ is unreliable near the edges of the bid support: the kernel window extends past the data.

Standard practice:

- Trim **pseudo-values ($\hat{v}$)** computed from bids near $\min_{i,t} b_{it}$ or $\max_{i,t} b_{it}$.
- Use a boundary-corrected ~~or higher-order~~ kernel in Step 1, since $\widehat{v}_{it}$ inherits the bias of $\widehat{g}$.

Conditioning on Auction Heterogeneity

Auctions often differ in observable characteristics X_t (number of bidders, item quality, location).

If bidders' values depend on X_t, estimate $G(\cdot \mid X_t)$ and $g(\cdot \mid X_t)$ by local (kernel-weighted) versions of Step 1, so that pseudo-values are computed within auctions with similar X_t.

This recovers $F(\cdot \mid X)$, the value distribution conditional on observables.

Why Nonparametric?

A parametric guess for F (e.g., Uniform, Lognormal) can be badly wrong, especially in the tails.

Because auction outcomes are determined by bidders with highest valuations (ie, upper tail),

counterfactual predictions — optimal reserve prices, expected revenue under a different mechanism — are often sensitive to exactly the part of F a misspecified parametric family gets wrong.

GPV avoids this: $\widehat{F}$ is built directly from data, with no functional-form assumption.

A Caveat: Everything Above Holds n Fixed

The bid distribution $G(\cdot)$ is not a primitive of the model — it is an output of the equilibrium strategy $\beta(\cdot; n)$, which itself depends on n.

More rivals $\Longrightarrow$ less shading $\Longrightarrow$ a different mapping from values to bids. So two auctions with the same underlying F but different n generate different bid distributions:

$$G(\cdot \mid n) \neq G(\cdot \mid n').$$

Practical consequence: Step 1 must estimate $\widehat{G}(\cdot \mid n)$, $\widehat{g}(\cdot \mid n)$ separately for each n, and pseudo-values must be constructed within each n-group before pooling.

First-Order Stochastic Dominance

Definition. Distribution A first-order stochastically dominates B if

$$F_A(x) \leq F_B(x) \qquad \text{for all } x.$$

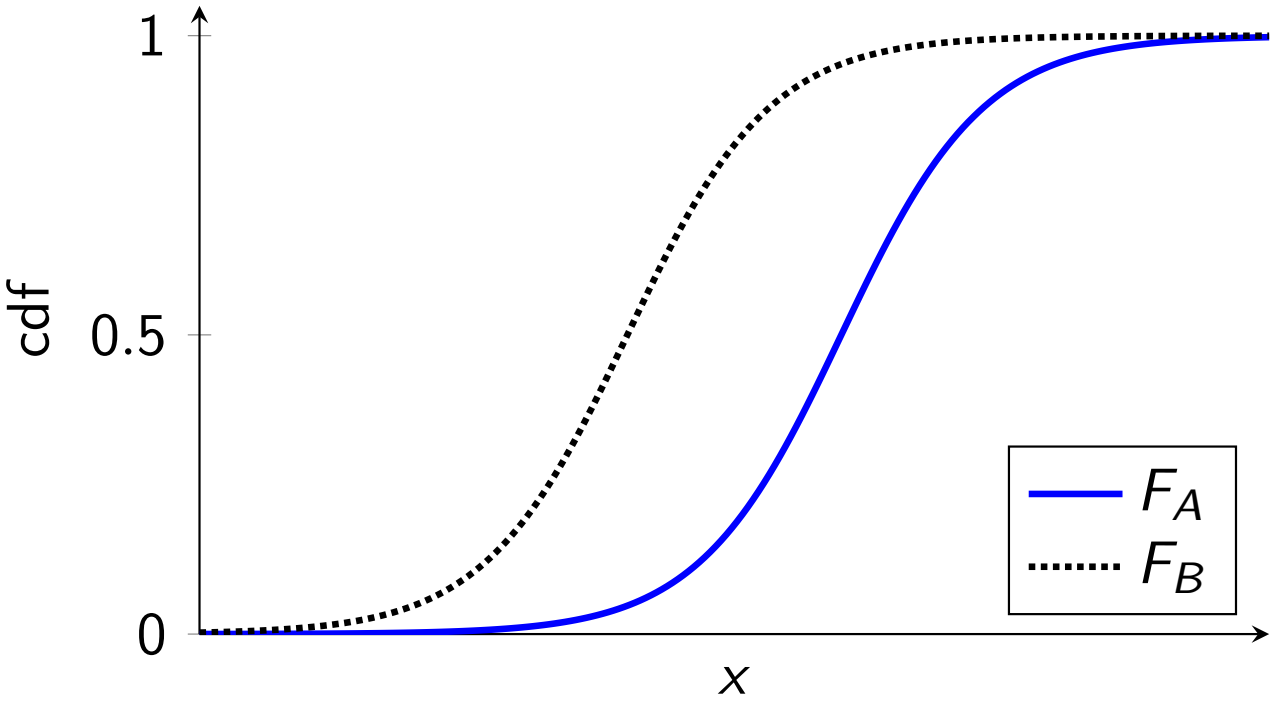
Equivalently, at every threshold x, A is at least as likely to clear it:

$$\Pr(A > x) \geq \Pr(B > x) \qquad \text{for all } x.$$

In words: A always puts more weight on higher outcomes. On a plot of the two cdfs, the dominant one lies to the right.

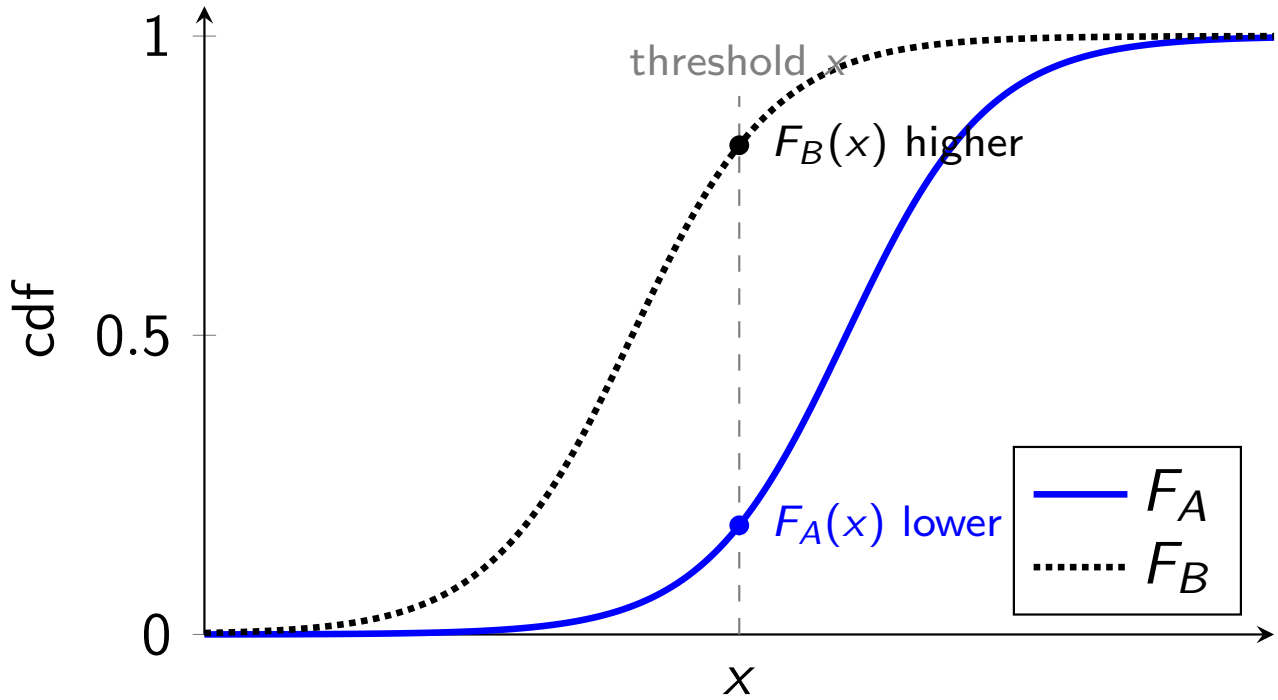
FOSD: A Picture

Here are two cdfs. Which one dominates the other?



FOSD: F_A Lies to the Right

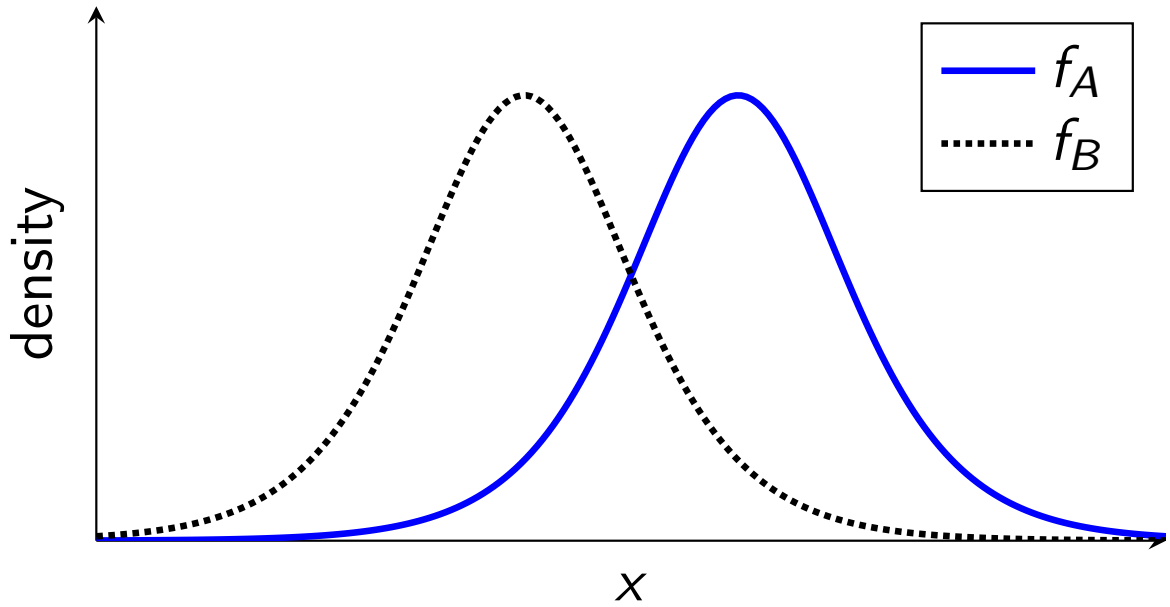
Pick any x. The dominated cdf is higher there; the dominant cdf is lower, i.e. to the right.



At this x, $F_A(x) \leq F_B(x)$. Because it holds for all x, A FOSD B: A is more likely to land on high outcomes.

The Same Thing, Seen in the Densities

Does the density picture help? A's mass sits to the right of B's: the whole distribution is shifted toward larger outcomes.



Take-away: under any "more is better" criterion (the mean, the median, any increasing payoff), the dominant distribution A never looks worse.

How the Bid Distributions Differ

Fix F. More rivals means less shading, so every value maps to a higher bid: for $n' > n$, $\beta(v; n') \geq \beta(v; n)$ for all v. The whole bid distribution shifts right:

$$G(\cdot \mid n') \text{ FOSD } G(\cdot \mid n), \qquad \text{i.e.} \qquad G(b \mid n') \leq G(b \mid n) \quad \text{for all } b.$$

The value distribution F dominates every bid distribution, and larger-n bids sit closer to it.

Why this matters for GPV: the bids you feed in look systematically different across n, so you estimate $\widehat{G}(\cdot \mid n), \widehat{g}(\cdot \mid n)$ within each n-group, yet the recovered primitive F is the same for every n.

The Magic of GPV

We never have to solve for $\beta(\cdot; n)$, or even know the functional form of the shading term

$$\frac{\int_0^v F^{n-1}(z)\,dz}{F^{n-1}(v)}.$$

The inversion formula

$$v = b + \frac{G(b \mid n)}{(n-1)\,g(b \mid n)}$$

uses only what shading leaves behind in the data — the observed bid distribution — never the shading rule itself.

So GPV recovers v from b auction-by-auction, n-group by n-group, without ever having to characterize equilibrium bidding ex ante. Whatever bidders' shading turned out to be, the data already encode it.

Part IV

Using the Identified F

Counterfactual Applications

Once $\widehat{F}$ (and $\widehat{f}$) are in hand:

- Estimate the optimal reserve price r^* solving $\psi(r) = 0$, where $\psi(v) \equiv v - \dfrac{1 - F(v)}{f(v)}$. (from Lecture 5)
- Predict revenue and bidder surplus under alternative auction rules (second-price, different n, a different reserve).
- Compare with monopolistic pricing
- **Important caveat**: if your bid data comes from auctions with a reserve price, the GPV formula changes.

Summary

- Bids are shaded, so the empirical bid distribution is not F.
- GPV inversion: $v = b + \dfrac{G(b)}{(n-1)g(b)}$, using only observable bid-distribution objects (b, G, g).
- The beauty: we never solve the game. β and F are never computed en route — GPV reads each value straight off the bid that shading produced.
- Two-step nonparametric estimator: estimate $(\widehat{G}, \widehat{g})$ from bids for each n pool, invert to pseudo-values $\widehat{v}_{it}$, then estimate $\widehat{F}$ from the pseudo-values.
- Boundary trimming and conditioning on observables X (and on n) are standard refinements.
- The identified $\widehat{F}$ enables counterfactual and optimal-mechanism analysis without parametric assumptions.